



Proof · Worksheet 1

Chapter 1: Proof. Answer all questions. Show your working clearly. Total: 72 marks. Time: 90 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each method of proof. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- In every proof, define any letters you use, and finish with a clear **concluding statement**.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Proof by deduction	19	
B Proof by exhaustion	19	
C Disproof by counterexample	12	
D Proof by contradiction	18	
Roar-level challenge	4	
Total	72	

Section A Proof by deduction

19 marks

1. Prove that the sum of any three consecutive integers is a multiple of 3.

[3]

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2. Prove that the product of any two odd numbers is odd.

[3]

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3. Prove that, for all real values of x ,

[3]

$$x^2 - 6x + 10 > 0.$$

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4. Prove that the difference between the squares of any two consecutive odd numbers is a multiple of 8. [4]

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5. (a) Prove that, for all positive values of x and y ,

[4]

$$\sqrt{xy} \leq \frac{x+y}{2}.$$

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(b) Prove by counterexample that the statement

[2]

$$\sqrt{xy} \leq \frac{x+y}{2} \text{ for all real values of } x \text{ and } y$$

is false.

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(Total for Question 5 is 6 marks)



Section B Proof by exhaustion

19 marks

6. Prove by exhaustion that $n^2 + n + 11$ is prime for all integers n such that $1 \leq n \leq 6$.

[2]

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7. Prove by exhaustion that every prime number between 20 and 40 can be written in the form $6k - 1$ or $6k + 1$, where k is a positive integer.

[3]

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8. Prove by exhaustion that the last digit of a square number can never be 2, 3, 7 or 8.

[3]

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9. Prove that the square of any integer can be written in the form $3k$ or $3k + 1$, where k is an integer. [4]

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10. The integer n is **not** a multiple of 3.

- (a) Prove by exhaustion that $n^2 - 1$ is a multiple of 3. [3]

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- (b) Hence, or otherwise, prove that if p is a prime number greater than 3, then $p^2 - 1$ is a multiple of 24. [4]

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(Total for Question 10 is 7 marks)

**Section C Disproof by counterexample**

12 marks

11. Show, by means of a counterexample, that the following statement is false.

[2]

“The sum of two prime numbers is always even.”

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12. Show, by means of a counterexample, that the following statement is false.

[2]

“ $(x+1)^2 > x^2$ for all real values of x .”

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13. Sam claims that $3^n + 2$ is prime for every positive integer n .

[2]

Show that Sam is wrong.

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14. Show, by means of a counterexample, that the following statement is false.

[2]

“If a and b are real numbers with $a > b$, then $a^2 > b^2$.”

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15. (a) A student claims that $n^2 - n + 41$ is prime for all positive integers n . [2]
Without testing every value from $n = 1$ upwards, show that this claim is false.

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- (b) Prove that the following statement is not always true. [2]
“If x and y are both irrational, then xy is irrational.”

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(Total for Question 15 is 4 marks)



Section D Proof by contradiction

18 marks

16. Prove by contradiction that there is no largest even integer.

[3]

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17. Prove by contradiction that, for an integer n , if n^2 is odd then n is odd.

[3]

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18. Prove by contradiction that there are no integers a and b such that

[3]

$$6a + 9b = 1.$$

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19. Prove by contradiction that there are no positive integers x and y such that

[4]

$$x^2 - y^2 = 1.$$

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20. Prove by contradiction that there are infinitely many prime numbers.

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(Total for Question 20 is 5 marks)



ROAR-LEVEL CHALLENGE

4 marks

21. Prove that, for every integer n , the number $n^3 - n$ is a multiple of 6.

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END OF WORKSHEET

Total: 72 marks